

Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

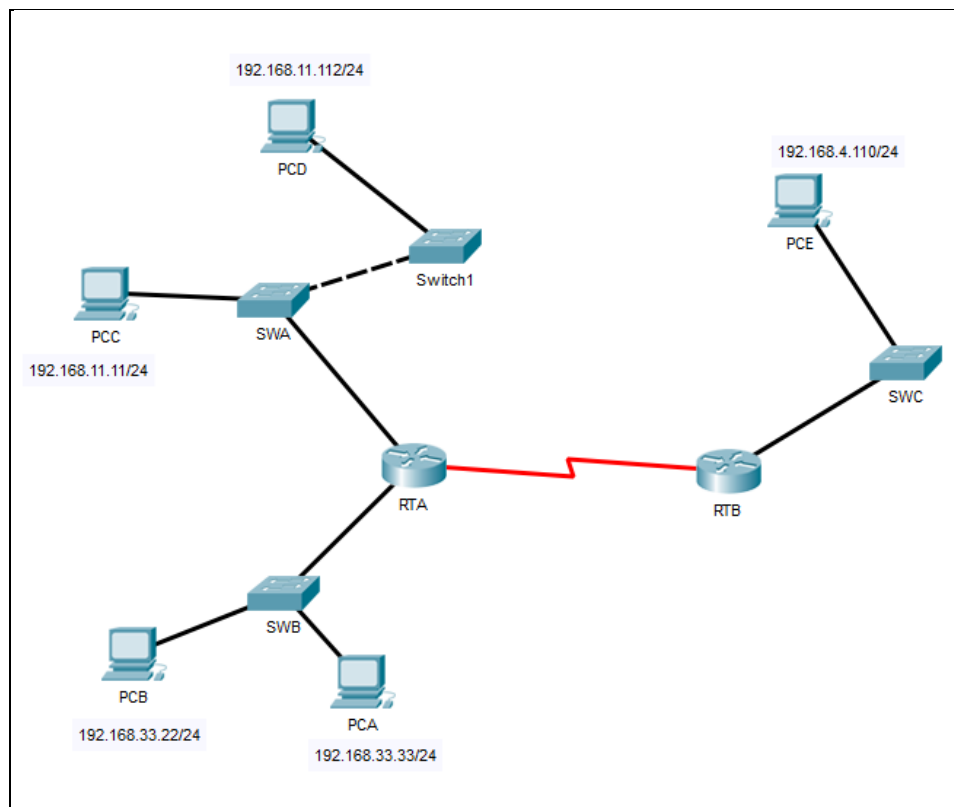


Figure 1

Part 1: Review the topology

Step 1: Perform the following tasks.

- At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.11.1 - 0002.4A00.0E91 ARPA FastEthernet1/0
Internet 192.168.33.1 - 000C.CF0C.593A ARPA FastEthernet0/0
RTA#
```

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.4.1 - 0001.977A.B614 ARPA FastEthernet0/0
RTB#
```

- c. At Switches SWA, SWB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

Switches SWA :

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
Mac Address Table
-----
Vlan Mac Address Type Ports
----
1 0002.4a00.0e91 DYNAMIC Fa0/1
1 000c.8546.7d85 DYNAMIC Fa1/1
SWA#
```

Switches SWB :

```
SWB>enable
SWB#show arp

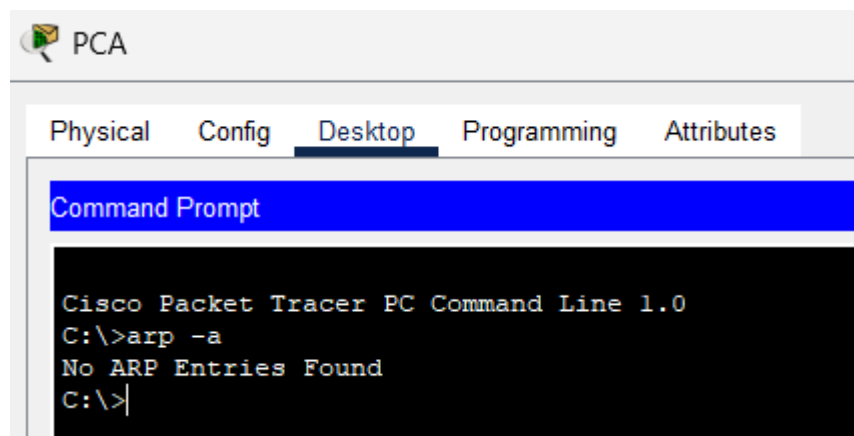
SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.cf0c.593a    DYNAMIC Fa0/1
SWB#
```

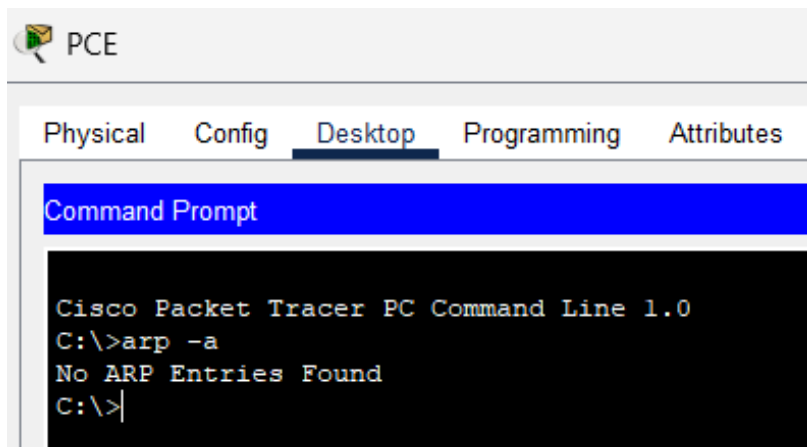
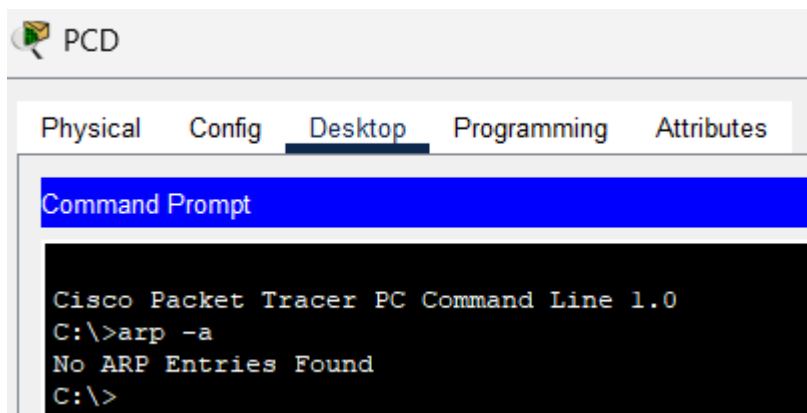
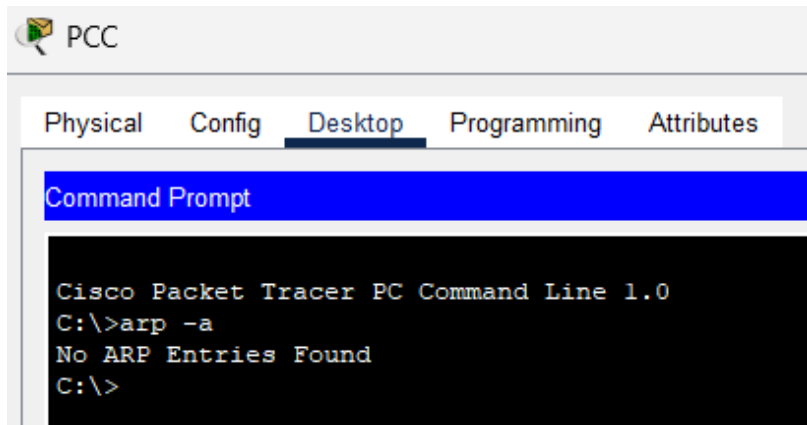
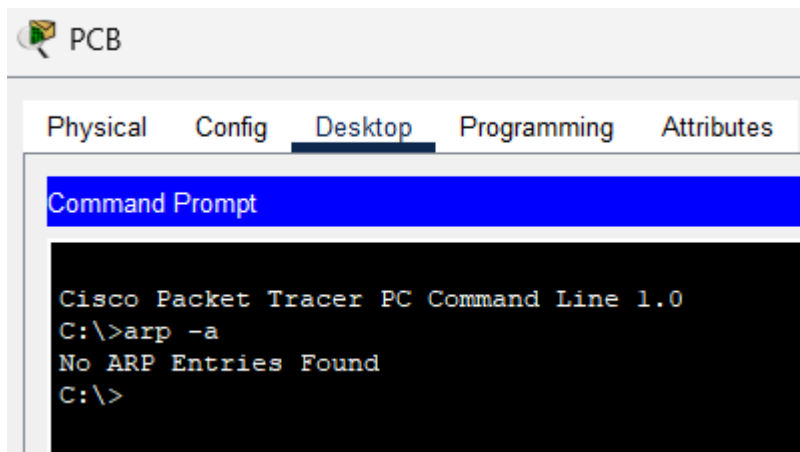
Switches SWC :

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.977a.b614    DYNAMIC Fa0/1
SWC#
```

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type `arp -a` and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.





- e. What are your thoughts on the results?

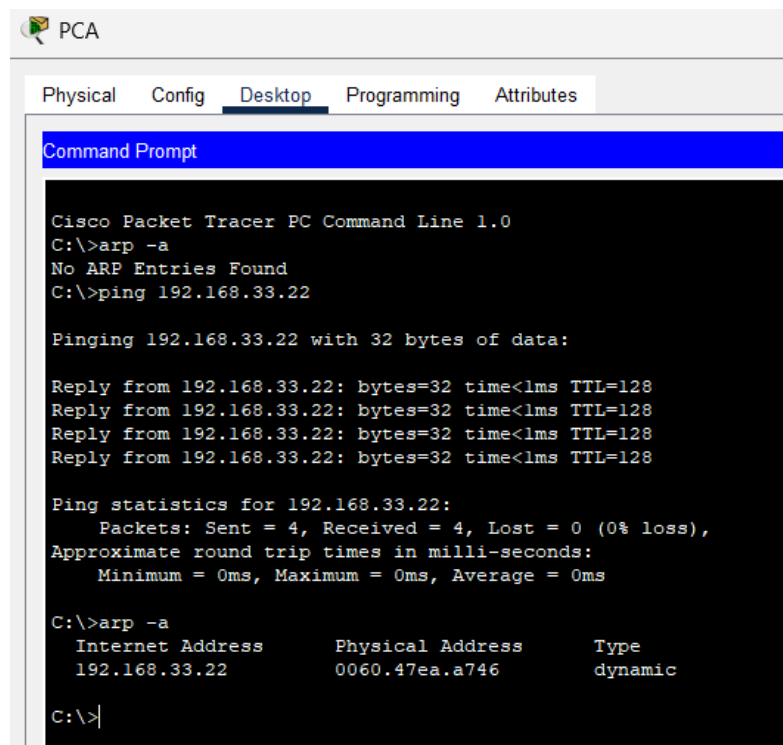
Ans : The PCs are not receiving any ARP request so there will be no entry in the PC ARP table.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- Click **PCA** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command. This may take a few seconds.
- In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

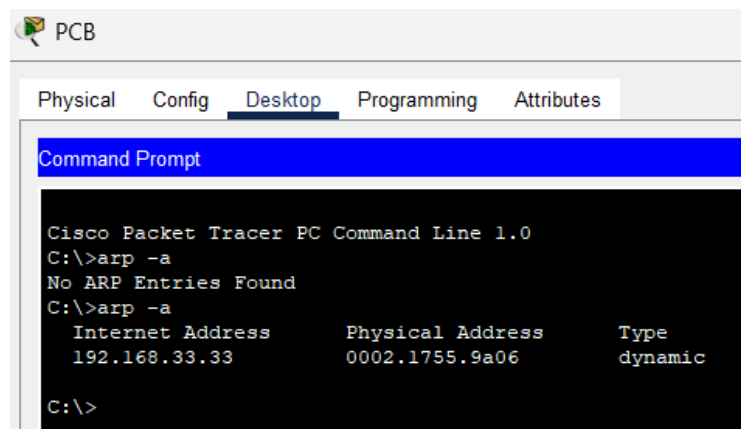
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>arp -a
    Internet Address      Physical Address        Type
    192.168.33.22         0060.47ea.a746         dynamic

C:\>
```

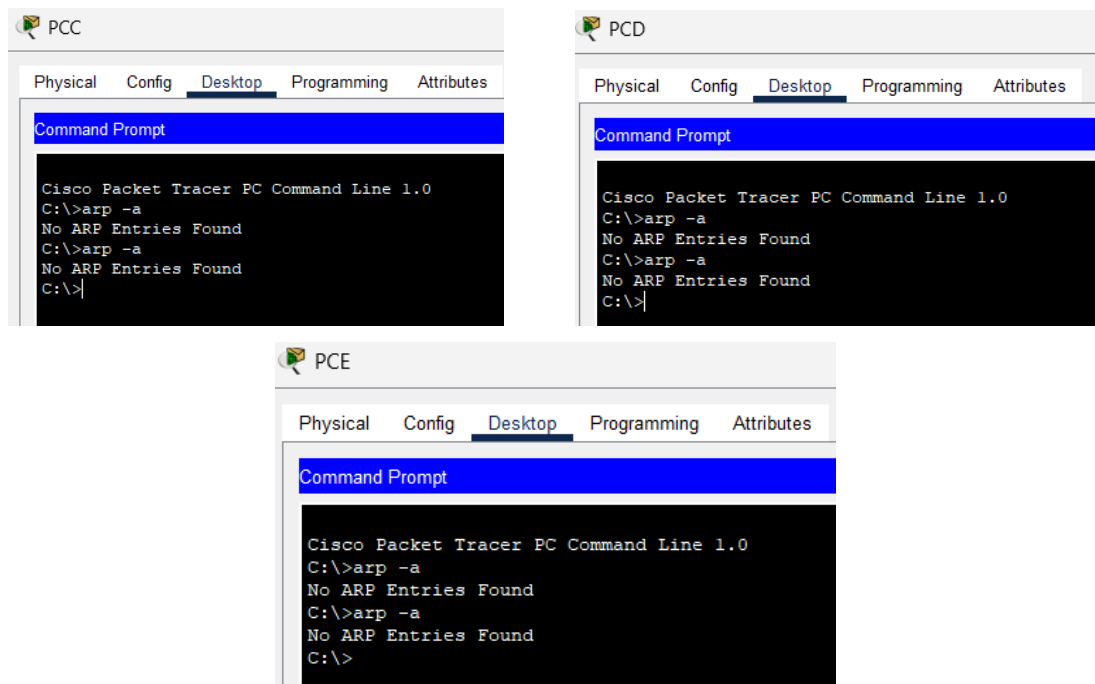
- In the Command prompt of PCB, type **arp -a**. Paste the result of this command here



```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
    Internet Address      Physical Address        Type
    192.168.33.33         0002.1755.9a06         dynamic

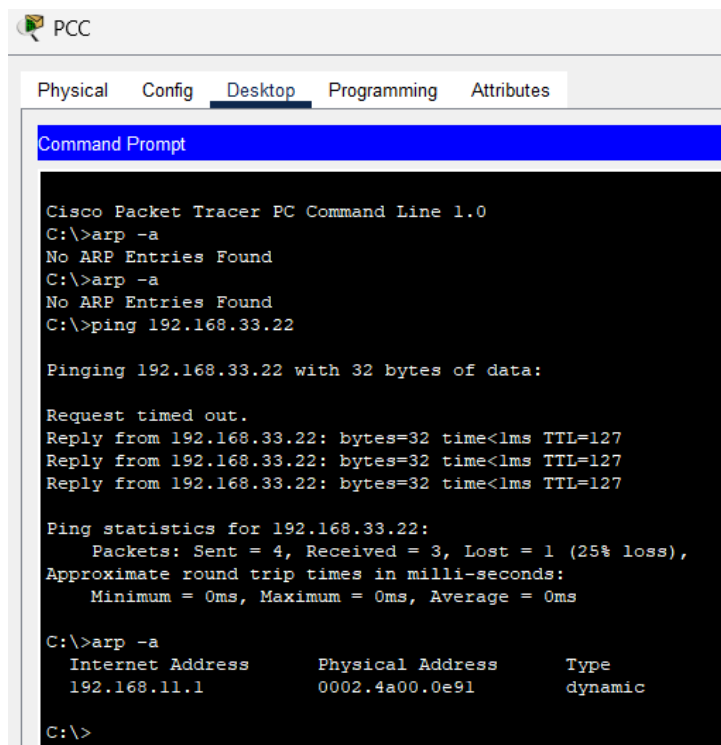
C:\>
```

- e. In the Command prompt of PCC, PCD and PCE, type **arp -a**. Paste the result of this command here.

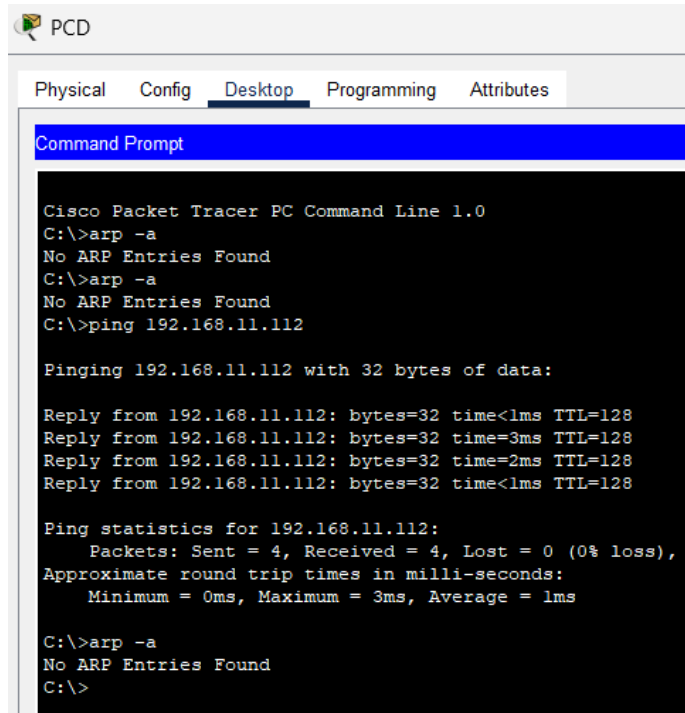


Step 2: Generate traffic between PCC to all other PC except PCA.

- Click **PCC** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.



- c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.



The screenshot shows the Cisco Packet Tracer interface for PC D. The 'Desktop' tab is selected, and the 'Command Prompt' window is open. The command prompt displays the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
C:\>ping 192.168.11.112

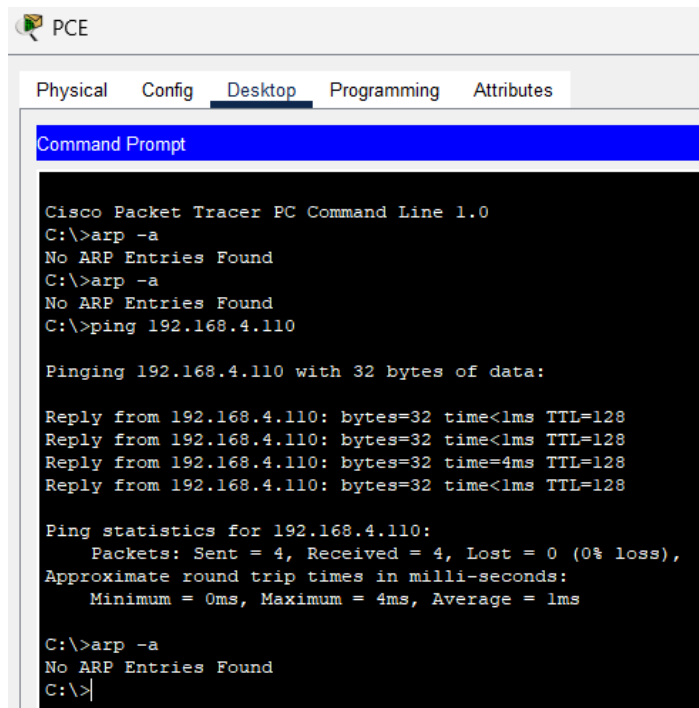
Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time=3ms TTL=128
Reply from 192.168.11.112: bytes=32 time=2ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>arp -a
No ARP Entries Found
C:\>
```

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.



The screenshot shows the Cisco Packet Tracer interface for PC E. The 'Desktop' tab is selected, and the 'Command Prompt' window is open. The command prompt displays the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>arp -a
No ARP Entries Found
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Reply from 192.168.4.110: bytes=32 time<1ms TTL=128
Reply from 192.168.4.110: bytes=32 time<1ms TTL=128
Reply from 192.168.4.110: bytes=32 time=4ms TTL=128
Reply from 192.168.4.110: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>arp -a
No ARP Entries Found
C:\>
```

- e. Discuss the results you got from all the commands on PCC.

Ans : I think from the results I got PCD and PCE did not ping successfully thus there is no ARP entries found from the command prompt.

- f. At Router RTA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.11.1      -          0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.11.11     7          00D0.D39A.C0D9  ARPA   FastEthernet1/0
Internet  192.168.33.1      -          000C.CF0C.593A  ARPA   FastEthernet0/0
Internet  192.168.33.22     7          0060.47EA.A746  ARPA   FastEthernet0/0
RTA#
```

- g. At Router RTA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.11.1      -          0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.11.11     7          00D0.D39A.C0D9  ARPA   FastEthernet1/0
Internet  192.168.33.1      -          000C.CF0C.593A  ARPA   FastEthernet0/0
Internet  192.168.33.22     7          0060.47EA.A746  ARPA   FastEthernet0/0
RTA#enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.11.1      -          0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.11.11     8          00D0.D39A.C0D9  ARPA   FastEthernet1/0
Internet  192.168.33.1      -          000C.CF0C.593A  ARPA   FastEthernet0/0
Internet  192.168.33.22     8          0060.47EA.A746  ARPA   FastEthernet0/0
RTA#
```


Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
      1    0002.4a00.0e91    DYNAMIC     Fa0/1
      1    000c.8546.7d85    DYNAMIC     Fa1/1
SWA#
```

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
      1    000c.cf0c.593a    DYNAMIC     Fa0/1
SWB#
```

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
      1    0001.977a.b614    DYNAMIC     Fa0/1
SWC#
```

- d. Do switches use arp table? (Y/N)

Ans : Yes

- e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetComm syllabi, it is just for knowledge..*

Ans : As the packet is being forwarded from one port to another, the switch will look up the IP address of the device in ARP table in order to obtain the MAC address.

- f. What information does the command `show mac-address-table` give?

Ans : It is used to display the information about the dynamic MAC address table entries.

Part 3: Attach wireless lab results.

In this part, you will use Lab 4 .pka file.

Step1: Change the filename of Lab 4.

- Change the Lab 4 filename to include your name. *Example: Lab4AliAhmad.pkt
- Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.

The screenshot displays the Cisco Packet Tracer interface. On the left, a network topology is shown with a central switch S1 connected to three PCs (PC1, PC2, PC3) and a wireless router WRS2. S1 has subinterfaces G0/0.10, G0/0.20, and G0/0.88. PC1 is on VLAN 10 (172.17.10.21/24), PC2 on VLAN 20 (172.17.20.22/24), and PC3 on VLAN 88 (172.17.40.100/24). WRS2 is on VLAN 88 (172.17.88.25/24). The PT Activity window on the right shows the 'Configuring Wireless LAN Access' task. It includes an 'Addressing Table' and a progress bar indicating 100% completion.

Device	Interface	IP Address	Subnet Mask	Default Gate
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1

- Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

This screenshot shows the Cisco Packet Tracer interface with a list of configuration steps on the left and the network topology on the right. The steps are organized into three parts: Part 1 (configure WRS2), Part 2 (configure PC3), and Part 3 (verify connectivity). The PT Activity window at the bottom shows the 'Configuring Wireless LAN Access' task with a progress bar indicating 100% completion.

Part 1: Configure WRS2 for wireless connectivity.

- At the top of the window, click **Wireless**. Set the **Network Mode** to **Wireless-N** Only and change the SSID to **WRS_LAN**.
- Disable **SSID Broadcast** and click **Save Settings**.
- Click the **Wireless Security** option.
- Change the **Security Mode** from **Disabled** to **WPA2 Personal**.
- Configure **crypto123** as the passphrase.
- Scroll to the bottom of the page and click **Save Settings**.

Part 2: Configure a Wireless Client

Step 1: Configure PC3 for wireless connectivity.

Because SSID broadcast is disabled, you must manually configure PC3 with the correct SSID and passphrase to establish a connection with the router.

- Click **PC3 > Desktop > PC Wireless**.
- Click the **Profiles** tab.
- Click **New**.
- Name the new profile **Wireless Access**.
- On the next screen, click **Advanced Setup**. Then manually enter the SSID of WRS_LAN on **Wireless Network Name**. Click **Next**.
- Choose **Obtain network settings automatically (DHCP)** as the network settings, and then click **Next**.
- On **Wireless Security**, choose **WPA2 Personal** as the method of encryption and click **Next**.
- Enter the passphrase **crypto123** and click **Next**.
- Click **Save** and then click **Connect to Network**.

Step 2: Verify PC3 wireless connectivity and IP addressing configuration.

- The **Signal Strength** and **Link Quality** indicators should show that you have a strong signal.
- Click **More Information** to see details of the connection including IP addressing information.
- Close the **PC Wireless** configuration window.

Part 3: Verify Connectivity

All the PCs should have connectivity with one another.